

Synapse Project Mastery Manual

This manual provides a comprehensive breakdown of the Synapse AI platform, mapping out the architecture, workflows, and implementation details of every feature to help you master the codebase.

Chapter 1: Authentication with Email Verification

Business Purpose & Workflow

This feature addresses specific student pain points by integrating AI with actionable insights. The workflow typically begins from the user interface, hits the Next.js API routes, passes through middleware, and is processed by the service layer before interacting with MongoDB.

Associated Files

- d:/Synapse/backend/src/controllers/authController.js
- d:/Synapse/backend/src/controllers/userController.js
- d:/Synapse/backend/src/controllers/usersController.js
- d:/Synapse/backend/src/middleware/auth.js
- d:/Synapse/backend/src/middleware/authMiddleware.js
- d:/Synapse/backend/src/middleware/verifyVaultAccess.js
- d:/Synapse/backend/src/models/User.js
- d:/Synapse/backend/src/routes/auth.js
- d:/Synapse/backend/src/routes/users.js
- d:/Synapse/frontend/src/assets/login_brain_split.png
- d:/Synapse/frontend/src/components/AuthLayout.css
- d:/Synapse/frontend/src/components/AuthLayout.jsx
- d:/Synapse/frontend/src/context/auth-context.js
- d:/Synapse/frontend/src/context/AuthContext.jsx
- d:/Synapse/frontend/src/hooks/useAuth.js
- d:/Synapse/frontend/src/pages/auth/ForgotPassword.jsx
- d:/Synapse/frontend/src/pages/auth/Login.jsx
- d:/Synapse/frontend/src/pages/auth/Onboarding.jsx
- d:/Synapse/frontend/src/pages/auth/Register.jsx
- d:/Synapse/frontend/src/pages/auth/ResetPassword.jsx
- d:/Synapse/frontend/src/pages/auth/VerifyEmail.jsx

Internal Implementation & Debugging

When debugging this feature, always check the associated controller and service files first. Ensure MongoDB collections are correctly structured and verify that API requests have the correct JWT tokens. Common issues arise from missing await statements or undefined context variables during AI calls.

Revision Checklist

- ☐ Understand the user journey for Authentication with Email Verification
- ☐ Review the related models and controllers
- ☐ Identify the specific AI prompts or logic used (if applicable)

Chapter 2: Student Profile and Onboarding Flow

Business Purpose & Workflow

This feature addresses specific student pain points by integrating AI with actionable insights. The workflow typically begins from the user interface, hits the Next.js API routes, passes through middleware, and is processed by the service layer before interacting with MongoDB.

Associated Files

- d:/Synapse/backend/src/routes/settings.js
- d:/Synapse/frontend/src/pages/auth/Onboarding.jsx
- d:/Synapse/frontend/src/pages/settings/AIPersonalizationSettings.jsx
- d:/Synapse/frontend/src/pages/settings/Settings.jsx

Internal Implementation & Debugging

When debugging this feature, always check the associated controller and service files first. Ensure MongoDB collections are correctly structured and verify that API requests have the correct JWT tokens. Common issues arise from missing await statements or undefined context variables during AI calls.

Revision Checklist

- ☐ Understand the user journey for Student Profile and Onboarding Flow
- ☐ Review the related models and controllers
- ☐ Identify the specific AI prompts or logic used (if applicable)

Chapter 3: Today Dashboard with AI Morning Briefing

Business Purpose & Workflow

This feature addresses specific student pain points by integrating AI with actionable insights. The workflow typically begins from the user interface, hits the Next.js API routes, passes through middleware, and is processed by the service layer before interacting with MongoDB.

Associated Files

- d:/Synapse/backend/src/models/Briefing.js
- d:/Synapse/frontend/src/pages/dashboard/TodayDashboard.jsx

Internal Implementation & Debugging

When debugging this feature, always check the associated controller and service files first. Ensure MongoDB collections are correctly structured and verify that API requests have the correct JWT tokens. Common issues arise from missing await statements or undefined context variables during AI calls.

Revision Checklist

- ☐ Understand the user journey for Today Dashboard with AI Morning Briefing
- ☐ Review the related models and controllers
- ☐ Identify the specific AI prompts or logic used (if applicable)

Chapter 4: Academic Tracker

Business Purpose & Workflow

This feature addresses specific student pain points by integrating AI with actionable insights. The workflow typically begins from the user interface, hits the Next.js API routes, passes through middleware, and is processed by the service layer before interacting with MongoDB.

Associated Files

- d:/Synapse/backend/src/controllers/academicsController.js
- d:/Synapse/backend/src/controllers/semesterController.js
- d:/Synapse/backend/src/controllers/subjectFileController.js
- d:/Synapse/backend/src/models/Attendance.js
- d:/Synapse/backend/src/models/Mark.js
- d:/Synapse/backend/src/models/Semester.js
- d:/Synapse/backend/src/models/Subject.js
- d:/Synapse/backend/src/models/SubjectFile.js
- d:/Synapse/backend/src/routes/academics.js
- d:/Synapse/backend/src/routes/semesters.js
- d:/Synapse/backend/src/routes/subjectFiles.js
- d:/Synapse/backend/src/services/academicService.js
- d:/Synapse/frontend/src/components/MarkdownText.jsx
- d:/Synapse/frontend/src/components/subjects/SubjectDrawer.jsx
- d:/Synapse/frontend/src/pages/academics/Academics.jsx
- d:/Synapse/frontend/src/pages/academics/AcademicTracker.jsx
- d:/Synapse/frontend/src/pages/academics/AddSemesterModal.jsx
- d:/Synapse/frontend/src/pages/academics/ExamImportModal.jsx
- d:/Synapse/frontend/src/pages/academics/ExamScheduleList.jsx
- d:/Synapse/frontend/src/pages/academics/ImportGradeCard.jsx
- d:/Synapse/frontend/src/pages/academics/MigrationModal.jsx
- d:/Synapse/frontend/src/pages/academics/SemesterList.jsx
- d:/Synapse/frontend/src/pages/academics/SemesterWorkspace.jsx
- d:/Synapse/frontend/src/pages/academics/Timetable.jsx
- d:/Synapse/frontend/src/pages/academics/TimetableImportModal.jsx
- d:/Synapse/frontend/src/pages/academics/WhatIfCalculator.jsx
- d:/Synapse/frontend/src/pages/subjects/Subjects.jsx
- d:/Synapse/frontend/src/services/subjectFileService.js

Internal Implementation & Debugging

When debugging this feature, always check the associated controller and service files first. Ensure MongoDB collections are correctly structured and verify that API requests have the correct JWT tokens. Common issues arise from missing await statements or undefined context variables during AI calls.

Revision Checklist

- ☐ Understand the user journey for Academic Tracker
- ☐ Review the related models and controllers
- ☐ Identify the specific AI prompts or logic used (if applicable)

Chapter 5: CGPA What-If Calculator

Business Purpose & Workflow

This feature addresses specific student pain points by integrating AI with actionable insights. The workflow typically begins from the user interface, hits the Next.js API routes, passes through middleware, and is processed by the service layer before interacting with MongoDB.

Associated Files

- d:/Synapse/frontend/src/pages/academics/WhatIfCalculator.jsx

Internal Implementation & Debugging

When debugging this feature, always check the associated controller and service files first. Ensure MongoDB collections are correctly structured and verify that API requests have the correct JWT tokens. Common issues arise from missing await statements or undefined context variables during AI calls.

Revision Checklist

- ☐ Understand the user journey for CGPA What-If Calculator
- ☐ Review the related models and controllers
- ☐ Identify the specific AI prompts or logic used (if applicable)

Chapter 6: AI Explains My Marks

Business Purpose & Workflow

This feature addresses specific student pain points by integrating AI with actionable insights. The workflow typically begins from the user interface, hits the Next.js API routes, passes through middleware, and is processed by the service layer before interacting with MongoDB.

Associated Files

- d:/Synapse/backend/src/controllers/aiController.js
- d:/Synapse/backend/src/routes/ai.js
- d:/Synapse/backend/src/services/aiContextBuilder.js
- d:/Synapse/backend/src/services/aiPreferenceNormalizer.js
- d:/Synapse/backend/src/services/aiRouter.js
- d:/Synapse/backend/src/utills/aiNotes.js
- d:/Synapse/backend/src/utills/groqGroupAI.js
- d:/Synapse/frontend/src/assets/brain_network.png
- d:/Synapse/frontend/src/assets/interactive_brain_split.png
- d:/Synapse/frontend/src/assets/login_brain_split.png
- d:/Synapse/frontend/src/components/ui/ai-accent.jsx
- d:/Synapse/frontend/src/main.jsx
- d:/Synapse/frontend/src/pages/auth/VerifyEmail.jsx
- d:/Synapse/frontend/src/pages/career/CareerDocDetailView.jsx
- d:/Synapse/frontend/src/pages/settings/AIPersonalizationSettings.jsx

Internal Implementation & Debugging

When debugging this feature, always check the associated controller and service files first. Ensure MongoDB collections are correctly structured and verify that API requests have the correct JWT tokens. Common issues arise from missing await statements or undefined context variables during AI calls.

Revision Checklist

- ☐ Understand the user journey for AI Explains My Marks
- ☐ Review the related models and controllers
- ☐ Identify the specific AI prompts or logic used (if applicable)

Chapter 7: AI Study Planner

Business Purpose & Workflow

This feature addresses specific student pain points by integrating AI with actionable insights. The workflow typically begins from the user interface, hits the Next.js API routes, passes through middleware, and is processed by the service layer before interacting with MongoDB.

Associated Files

- d:/Synapse/backend/src/controllers/plannerController.js
- d:/Synapse/backend/src/models/ExamSchedule.js
- d:/Synapse/backend/src/models/StudyTask.js
- d:/Synapse/backend/src/routes/planner.js
- d:/Synapse/backend/src/services/studyPlannerService.js
- d:/Synapse/backend/src/utils/plannerScheduler.js
- d:/Synapse/frontend/src/components/planner/CustomPdfPlanModal.jsx
- d:/Synapse/frontend/src/pages/academics/ExamScheduleList.jsx
- d:/Synapse/frontend/src/pages/planner/FocusMode.jsx
- d:/Synapse/frontend/src/pages/planner/Planner.jsx

Internal Implementation & Debugging

When debugging this feature, always check the associated controller and service files first. Ensure MongoDB collections are correctly structured and verify that API requests have the correct JWT tokens. Common issues arise from missing await statements or undefined context variables during AI calls.

Revision Checklist

- ☐ Understand the user journey for AI Study Planner
- ☐ Review the related models and controllers
- ☐ Identify the specific AI prompts or logic used (if applicable)

Chapter 8: Focus Mode - Pomodoro Study Timer

Business Purpose & Workflow

This feature addresses specific student pain points by integrating AI with actionable insights. The workflow typically begins from the user interface, hits the Next.js API routes, passes through middleware, and is processed by the service layer before interacting with MongoDB.

Associated Files

- d:/Synapse/backend/src/controllers/focusController.js
- d:/Synapse/backend/src/models/FocusSession.js
- d:/Synapse/backend/src/routes/focus.js
- d:/Synapse/frontend/src/context/FocusTimerContext.jsx
- d:/Synapse/frontend/src/pages/planner/FocusMode.jsx

Internal Implementation & Debugging

When debugging this feature, always check the associated controller and service files first. Ensure MongoDB collections are correctly structured and verify that API requests have the correct JWT tokens. Common issues arise from missing await statements or undefined context variables during AI calls.

Revision Checklist

- ☐ Understand the user journey for Focus Mode - Pomodoro Study Timer
- ☐ Review the related models and controllers
- ☐ Identify the specific AI prompts or logic used (if applicable)

Chapter 9: AI Notebook

Business Purpose & Workflow

This feature addresses specific student pain points by integrating AI with actionable insights. The workflow typically begins from the user interface, hits the Next.js API routes, passes through middleware, and is processed by the service layer before interacting with MongoDB.

Associated Files

- d:/Synapse/backend/src/controllers/notebookController.js
- d:/Synapse/backend/src/models/Notebook.js
- d:/Synapse/backend/src/models/NotebookChat.js
- d:/Synapse/backend/src/models/SavedSummary.js
- d:/Synapse/backend/src/routes/notebook.js
- d:/Synapse/backend/src/utils/aiNotes.js
- d:/Synapse/backend/src/utils/generateSummaryPdf.js
- d:/Synapse/frontend/src/pages/notebook/Notebook.jsx
- d:/Synapse/frontend/src/pages/notebook/NotebookList.jsx
- d:/Synapse/frontend/src/pages/notebook/NotebookView.jsx

Internal Implementation & Debugging

When debugging this feature, always check the associated controller and service files first. Ensure MongoDB collections are correctly structured and verify that API requests have the correct JWT tokens. Common issues arise from missing await statements or undefined context variables during AI calls.

Revision Checklist

- ☐ Understand the user journey for AI Notebook
- ☐ Review the related models and controllers
- ☐ Identify the specific AI prompts or logic used (if applicable)

Chapter 10: Finance Tracker

Business Purpose & Workflow

This feature addresses specific student pain points by integrating AI with actionable insights. The workflow typically begins from the user interface, hits the Next.js API routes, passes through middleware, and is processed by the service layer before interacting with MongoDB.

Associated Files

- d:/Synapse/backend/src/controllers/financeController.js
- d:/Synapse/backend/src/jobs/recurringExpenses.js
- d:/Synapse/backend/src/models/Budget.js
- d:/Synapse/backend/src/models/Expense.js
- d:/Synapse/backend/src/routes/finance.js
- d:/Synapse/frontend/src/pages/finance/Finance.jsx
- d:/Synapse/frontend/src/pages/finance/FinanceTracker.jsx

Internal Implementation & Debugging

When debugging this feature, always check the associated controller and service files first. Ensure MongoDB collections are correctly structured and verify that API requests have the correct JWT tokens. Common issues arise from missing await statements or undefined context variables during AI calls.

Revision Checklist

- ☐ Understand the user journey for Finance Tracker
- ☐ Review the related models and controllers
- ☐ Identify the specific AI prompts or logic used (if applicable)

Chapter 11: Habit Tracker

Business Purpose & Workflow

This feature addresses specific student pain points by integrating AI with actionable insights. The workflow typically begins from the user interface, hits the Next.js API routes, passes through middleware, and is processed by the service layer before interacting with MongoDB.

Associated Files

- d:/Synapse/backend/src/controllers/habitsController.js
- d:/Synapse/backend/src/jobs/habitReminders.js
- d:/Synapse/backend/src/models/Habit.js
- d:/Synapse/backend/src/models/HabitLog.js
- d:/Synapse/backend/src/routes/habits.js
- d:/Synapse/frontend/src/pages/academics/AcademicTracker.jsx
- d:/Synapse/frontend/src/pages/finance/FinanceTracker.jsx
- d:/Synapse/frontend/src/pages/habits/Habits.jsx
- d:/Synapse/frontend/src/pages/habits/HabitTracker.jsx

Internal Implementation & Debugging

When debugging this feature, always check the associated controller and service files first. Ensure MongoDB collections are correctly structured and verify that API requests have the correct JWT tokens. Common issues arise from missing await statements or undefined context variables during AI calls.

Revision Checklist

- ☐ Understand the user journey for Habit Tracker
- ☐ Review the related models and controllers
- ☐ Identify the specific AI prompts or logic used (if applicable)

Chapter 12: Central AI Assistant

Business Purpose & Workflow

This feature addresses specific student pain points by integrating AI with actionable insights. The workflow typically begins from the user interface, hits the Next.js API routes, passes through middleware, and is processed by the service layer before interacting with MongoDB.

Associated Files

- d:/Synapse/backend/src/controllers/aiController.js
- d:/Synapse/backend/src/models/NotebookChat.js
- d:/Synapse/backend/src/routes/ai.js
- d:/Synapse/backend/src/services/aiContextBuilder.js
- d:/Synapse/backend/src/services/aiPreferenceNormalizer.js
- d:/Synapse/backend/src/services/aiRouter.js
- d:/Synapse/backend/src/utils/aiNotes.js
- d:/Synapse/backend/src/utils/groqGroupAI.js
- d:/Synapse/frontend/src/assets/brain_network.png
- d:/Synapse/frontend/src/assets/interactive_brain_split.png
- d:/Synapse/frontend/src/assets/login_brain_split.png
- d:/Synapse/frontend/src/components/ChatWidget.jsx
- d:/Synapse/frontend/src/components/messaging/ChatInput.jsx
- d:/Synapse/frontend/src/components/messaging/ChatWindow.jsx
- d:/Synapse/frontend/src/components/ui/ai-accent.jsx
- d:/Synapse/frontend/src/components/ui/ChatBubble.jsx
- d:/Synapse/frontend/src/main.jsx
- d:/Synapse/frontend/src/pages/auth/VerifyEmail.jsx
- d:/Synapse/frontend/src/pages/career/CareerDocDetailView.jsx
- d:/Synapse/frontend/src/pages/groups/GroupChat.jsx
- d:/Synapse/frontend/src/pages/resources/ExplorerChat.jsx
- d:/Synapse/frontend/src/pages/settings/AIPersonalizationSettings.jsx

Internal Implementation & Debugging

When debugging this feature, always check the associated controller and service files first. Ensure MongoDB collections are correctly structured and verify that API requests have the correct JWT tokens. Common issues arise from missing await statements or undefined context variables during AI calls.

Revision Checklist

- ☐ Understand the user journey for Central AI Assistant
- ☐ Review the related models and controllers
- ☐ Identify the specific AI prompts or logic used (if applicable)

Chapter 13: Weekly AI Report Card

Business Purpose & Workflow

This feature addresses specific student pain points by integrating AI with actionable insights. The workflow typically begins from the user interface, hits the Next.js API routes, passes through middleware, and is processed by the service layer before interacting with MongoDB.

Associated Files

- d:/Synapse/backend/src/jobs/weeklyReport.js
- d:/Synapse/backend/src/models/WeeklyReport.js

Internal Implementation & Debugging

When debugging this feature, always check the associated controller and service files first. Ensure MongoDB collections are correctly structured and verify that API requests have the correct JWT tokens. Common issues arise from missing await statements or undefined context variables during AI calls.

Revision Checklist

- ☐ Understand the user journey for Weekly AI Report Card
- ☐ Review the related models and controllers
- ☐ Identify the specific AI prompts or logic used (if applicable)

Chapter 14: Study Groups

Business Purpose & Workflow

This feature addresses specific student pain points by integrating AI with actionable insights. The workflow typically begins from the user interface, hits the Next.js API routes, passes through middleware, and is processed by the service layer before interacting with MongoDB.

Associated Files

- d:/Synapse/backend/src/controllers/groupsController.js
- d:/Synapse/backend/src/models/GroupMessage.js
- d:/Synapse/backend/src/models/StudyGroup.js
- d:/Synapse/backend/src/routes/groups.js
- d:/Synapse/backend/src/utls/groqGroupAI.js
- d:/Synapse/backend/src/utls/groupPermissions.js
- d:/Synapse/backend/src/utls/saveGroupBotMessage.js
- d:/Synapse/frontend/src/pages/groups/GroupChat.jsx
- d:/Synapse/frontend/src/pages/groups/GroupInfoPanel.jsx
- d:/Synapse/frontend/src/pages/groups/GroupPermissionsPanel.jsx
- d:/Synapse/frontend/src/pages/groups/Groups.jsx
- d:/Synapse/frontend/src/pages/groups/StudyGroups.jsx

Internal Implementation & Debugging

When debugging this feature, always check the associated controller and service files first. Ensure MongoDB collections are correctly structured and verify that API requests have the correct JWT tokens. Common issues arise from missing await statements or undefined context variables during AI calls.

Revision Checklist

- ☐ Understand the user journey for Study Groups
- ☐ Review the related models and controllers
- ☐ Identify the specific AI prompts or logic used (if applicable)

Chapter 15: Quick Capture

Business Purpose & Workflow

This feature addresses specific student pain points by integrating AI with actionable insights. The workflow typically begins from the user interface, hits the Next.js API routes, passes through middleware, and is processed by the service layer before interacting with MongoDB.

Associated Files

- d:/Synapse/frontend/src/components/QuickCapture.jsx

Internal Implementation & Debugging

When debugging this feature, always check the associated controller and service files first. Ensure MongoDB collections are correctly structured and verify that API requests have the correct JWT tokens. Common issues arise from missing await statements or undefined context variables during AI calls.

Revision Checklist

- ☐ Understand the user journey for Quick Capture
- ☐ Review the related models and controllers
- ☐ Identify the specific AI prompts or logic used (if applicable)

Chapter 16: Notifications System

Business Purpose & Workflow

This feature addresses specific student pain points by integrating AI with actionable insights. The workflow typically begins from the user interface, hits the Next.js API routes, passes through middleware, and is processed by the service layer before interacting with MongoDB.

Associated Files

- d:/Synapse/backend/src/controllers/notificationsController.js
- d:/Synapse/backend/src/models/Notification.js
- d:/Synapse/backend/src/routes/notifications.js
- d:/Synapse/backend/src/services/notificationService.js
- d:/Synapse/frontend/src/components/NotificationBell.jsx
- d:/Synapse/frontend/src/components/NotificationsPopup.jsx
- d:/Synapse/frontend/src/pages/notifications/NotificationsList.jsx

Internal Implementation & Debugging

When debugging this feature, always check the associated controller and service files first. Ensure MongoDB collections are correctly structured and verify that API requests have the correct JWT tokens. Common issues arise from missing await statements or undefined context variables during AI calls.

Revision Checklist

- ☐ Understand the user journey for Notifications System
- ☐ Review the related models and controllers
- ☐ Identify the specific AI prompts or logic used (if applicable)

Chapter 17: Settings

Business Purpose & Workflow

This feature addresses specific student pain points by integrating AI with actionable insights. The workflow typically begins from the user interface, hits the Next.js API routes, passes through middleware, and is processed by the service layer before interacting with MongoDB.

Associated Files

- d:/Synapse/backend/src/routes/settings.js
- d:/Synapse/frontend/src/pages/settings/AIPersonalizationSettings.jsx
- d:/Synapse/frontend/src/pages/settings/Settings.jsx

Internal Implementation & Debugging

When debugging this feature, always check the associated controller and service files first. Ensure MongoDB collections are correctly structured and verify that API requests have the correct JWT tokens. Common issues arise from missing await statements or undefined context variables during AI calls.

Revision Checklist

- ☐ Understand the user journey for Settings
- ☐ Review the related models and controllers
- ☐ Identify the specific AI prompts or logic used (if applicable)

Chapter 18: AI Learning Resource Explorer

Business Purpose & Workflow

This feature addresses specific student pain points by integrating AI with actionable insights. The workflow typically begins from the user interface, hits the Next.js API routes, passes through middleware, and is processed by the service layer before interacting with MongoDB.

Associated Files

- d:/Synapse/backend/src/controllers/resourceController.js
- d:/Synapse/backend/src/models/ExplorerSession.js
- d:/Synapse/backend/src/routes/resources.js
- d:/Synapse/frontend/src/pages/resources/ExplorerChat.jsx
- d:/Synapse/frontend/src/pages/resources/ResourceExplorer.jsx

Internal Implementation & Debugging

When debugging this feature, always check the associated controller and service files first. Ensure MongoDB collections are correctly structured and verify that API requests have the correct JWT tokens. Common issues arise from missing await statements or undefined context variables during AI calls.

Revision Checklist

- ☐ Understand the user journey for AI Learning Resource Explorer
- ☐ Review the related models and controllers
- ☐ Identify the specific AI prompts or logic used (if applicable)

Chapter 19: Smart Task Prioritisation

Business Purpose & Workflow

This feature addresses specific student pain points by integrating AI with actionable insights. The workflow typically begins from the user interface, hits the Next.js API routes, passes through middleware, and is processed by the service layer before interacting with MongoDB.

Associated Files

- d:/Synapse/backend/src/models/StudyTask.js

Internal Implementation & Debugging

When debugging this feature, always check the associated controller and service files first. Ensure MongoDB collections are correctly structured and verify that API requests have the correct JWT tokens. Common issues arise from missing await statements or undefined context variables during AI calls.

Revision Checklist

- ☐ Understand the user journey for Smart Task Prioritisation
- ☐ Review the related models and controllers
- ☐ Identify the specific AI prompts or logic used (if applicable)

Chapter 20: Study Analytics Dashboard

Business Purpose & Workflow

This feature addresses specific student pain points by integrating AI with actionable insights. The workflow typically begins from the user interface, hits the Next.js API routes, passes through middleware, and is processed by the service layer before interacting with MongoDB.

Associated Files

Files dynamically shared across modules or handled in core components.

Internal Implementation & Debugging

When debugging this feature, always check the associated controller and service files first. Ensure MongoDB collections are correctly structured and verify that API requests have the correct JWT tokens. Common issues arise from missing await statements or undefined context variables during AI calls.

Revision Checklist

- ☐ Understand the user journey for Study Analytics Dashboard
- ☐ Review the related models and controllers
- ☐ Identify the specific AI prompts or logic used (if applicable)

Chapter 21: Resource Recommendation Engine

Business Purpose & Workflow

This feature addresses specific student pain points by integrating AI with actionable insights. The workflow typically begins from the user interface, hits the Next.js API routes, passes through middleware, and is processed by the service layer before interacting with MongoDB.

Associated Files

Files dynamically shared across modules or handled in core components.

Internal Implementation & Debugging

When debugging this feature, always check the associated controller and service files first. Ensure MongoDB collections are correctly structured and verify that API requests have the correct JWT tokens. Common issues arise from missing await statements or undefined context variables during AI calls.

Revision Checklist

- ☐ Understand the user journey for Resource Recommendation Engine
- ☐ Review the related models and controllers
- ☐ Identify the specific AI prompts or logic used (if applicable)

Chapter 22: Exam Readiness Score

Business Purpose & Workflow

This feature addresses specific student pain points by integrating AI with actionable insights. The workflow typically begins from the user interface, hits the Next.js API routes, passes through middleware, and is processed by the service layer before interacting with MongoDB.

Associated Files

- d:/Synapse/backend/src/services/readinessService.js

Internal Implementation & Debugging

When debugging this feature, always check the associated controller and service files first. Ensure MongoDB collections are correctly structured and verify that API requests have the correct JWT tokens. Common issues arise from missing await statements or undefined context variables during AI calls.

Revision Checklist

- ☐ Understand the user journey for Exam Readiness Score
- ☐ Review the related models and controllers
- ☐ Identify the specific AI prompts or logic used (if applicable)

Chapter 23: Semester Performance Predictor

Business Purpose & Workflow

This feature addresses specific student pain points by integrating AI with actionable insights. The workflow typically begins from the user interface, hits the Next.js API routes, passes through middleware, and is processed by the service layer before interacting with MongoDB.

Associated Files

Files dynamically shared across modules or handled in core components.

Internal Implementation & Debugging

When debugging this feature, always check the associated controller and service files first. Ensure MongoDB collections are correctly structured and verify that API requests have the correct JWT tokens. Common issues arise from missing await statements or undefined context variables during AI calls.

Revision Checklist

- ☐ Understand the user journey for Semester Performance Predictor
- ☐ Review the related models and controllers
- ☐ Identify the specific AI prompts or logic used (if applicable)

Chapter 24: Universal Academic Calendar

Business Purpose & Workflow

This feature addresses specific student pain points by integrating AI with actionable insights. The workflow typically begins from the user interface, hits the Next.js API routes, passes through middleware, and is processed by the service layer before interacting with MongoDB.

Associated Files

- `d:/Synapse/backend/src/controllers/calendarController.js`
- `d:/Synapse/backend/src/jobs/eventReminders.js`
- `d:/Synapse/backend/src/models/CustomEvent.js`
- `d:/Synapse/backend/src/routes/calendar.js`
- `d:/Synapse/frontend/src/pages/calendar/Calendar.jsx`

Internal Implementation & Debugging

When debugging this feature, always check the associated controller and service files first. Ensure MongoDB collections are correctly structured and verify that API requests have the correct JWT tokens. Common issues arise from missing await statements or undefined context variables during AI calls.

Revision Checklist

- ☐ Understand the user journey for Universal Academic Calendar
- ☐ Review the related models and controllers
- ☐ Identify the specific AI prompts or logic used (if applicable)

Chapter 25: Peer Networking & Direct Messaging

Business Purpose & Workflow

This feature addresses specific student pain points by integrating AI with actionable insights. The workflow typically begins from the user interface, hits the Next.js API routes, passes through middleware, and is processed by the service layer before interacting with MongoDB.

Associated Files

- `d:/Synapse/backend/src/controllers/friendsController.js`
- `d:/Synapse/backend/src/models/Friendship.js`
- `d:/Synapse/backend/src/models/GroupMessage.js`
- `d:/Synapse/backend/src/models/Message.js`
- `d:/Synapse/backend/src/routes/friends.js`
- `d:/Synapse/backend/src/utils/saveGroupBotMessage.js`
- `d:/Synapse/frontend/src/assets/brain_network.png`
- `d:/Synapse/frontend/src/components/MessagesPopup.jsx`
- `d:/Synapse/frontend/src/components/messaging/MessageBubble.jsx`
- `d:/Synapse/frontend/src/pages/messages/Messages.jsx`

Internal Implementation & Debugging

When debugging this feature, always check the associated controller and service files first. Ensure MongoDB collections are correctly structured and verify that API requests have the correct JWT tokens. Common issues arise from missing await statements or undefined context variables during AI calls.

Revision Checklist

- ☐ Understand the user journey for Peer Networking & Direct Messaging
- ☐ Review the related models and controllers
- ☐ Identify the specific AI prompts or logic used (if applicable)

Chapter 26: Career Vault Security

Business Purpose & Workflow

This feature addresses specific student pain points by integrating AI with actionable insights. The workflow typically begins from the user interface, hits the Next.js API routes, passes through middleware, and is processed by the service layer before interacting with MongoDB.

Associated Files

- `d:/Synapse/backend/src/controllers/careerVaultController.js`
- `d:/Synapse/backend/src/middleware/careerVaultUpload.js`
- `d:/Synapse/backend/src/middleware/verifyVaultAccess.js`
- `d:/Synapse/backend/src/routes/careerVaultRoutes.js`
- `d:/Synapse/frontend/src/pages/career/CareerVaultGate.jsx`
- `d:/Synapse/frontend/src/pages/career/CareerVaultLayout.jsx`
- `d:/Synapse/frontend/src/pages/career/CareerVaultList.jsx`
- `d:/Synapse/frontend/src/pages/career/CareerVaultNav.jsx`
- `d:/Synapse/frontend/src/pages/career/ForgotVaultPassword.jsx`
- `d:/Synapse/frontend/src/pages/career/ResetVaultPassword.jsx`

Internal Implementation & Debugging

When debugging this feature, always check the associated controller and service files first. Ensure MongoDB collections are correctly structured and verify that API requests have the correct JWT tokens. Common issues arise from missing await statements or undefined context variables during AI calls.

Revision Checklist

- ☐ Understand the user journey for Career Vault Security
- ☐ Review the related models and controllers
- ☐ Identify the specific AI prompts or logic used (if applicable)

Chapter 27: Career Timeline

Business Purpose & Workflow

This feature addresses specific student pain points by integrating AI with actionable insights. The workflow typically begins from the user interface, hits the Next.js API routes, passes through middleware, and is processed by the service layer before interacting with MongoDB.

Associated Files

- d:/Synapse/backend/src/controllers/careerVaultController.js
- d:/Synapse/backend/src/middleware/careerVaultUpload.js
- d:/Synapse/backend/src/models/CareerDocument.js
- d:/Synapse/backend/src/routes/careerVaultRoutes.js
- d:/Synapse/backend/src/services/careerDocExtractor.js
- d:/Synapse/frontend/src/pages/career/CareerDocDetailView.jsx
- d:/Synapse/frontend/src/pages/career/CareerDocUploadModal.jsx
- d:/Synapse/frontend/src/pages/career/CareerTimeline.jsx
- d:/Synapse/frontend/src/pages/career/CareerVaultGate.jsx
- d:/Synapse/frontend/src/pages/career/CareerVaultLayout.jsx
- d:/Synapse/frontend/src/pages/career/CareerVaultList.jsx
- d:/Synapse/frontend/src/pages/career/CareerVaultNav.jsx
- d:/Synapse/frontend/src/pages/career/ForgotVaultPassword.jsx
- d:/Synapse/frontend/src/pages/career/ResetVaultPassword.jsx
- d:/Synapse/frontend/src/pages/career/ResumeBuilderList.jsx
- d:/Synapse/frontend/src/pages/career/ResumeEditor.jsx
- d:/Synapse/frontend/src/pages/career/ResumeIntelligence.jsx

Internal Implementation & Debugging

When debugging this feature, always check the associated controller and service files first. Ensure MongoDB collections are correctly structured and verify that API requests have the correct JWT tokens. Common issues arise from missing await statements or undefined context variables during AI calls.

Revision Checklist

- ☐ Understand the user journey for Career Timeline
- ☐ Review the related models and controllers
- ☐ Identify the specific AI prompts or logic used (if applicable)

Chapter 28: Resume Intelligence

Business Purpose & Workflow

This feature addresses specific student pain points by integrating AI with actionable insights. The workflow typically begins from the user interface, hits the Next.js API routes, passes through middleware, and is processed by the service layer before interacting with MongoDB.

Associated Files

- d:/Synapse/backend/src/controllers/resumeController.js
- d:/Synapse/backend/src/middleware/resumeUpload.js
- d:/Synapse/backend/src/models/Resume.js
- d:/Synapse/backend/src/services/resumeGenerator.js
- d:/Synapse/backend/src/services/resumeParser.js
- d:/Synapse/backend/src/services/resumePdfExporter.js
- d:/Synapse/backend/src/services/resumeRewriteService.js
- d:/Synapse/backend/src/services/resumeSkillGapAnalyzer.js
- d:/Synapse/backend/src/services/resumeTemplates/atsClassic.js
- d:/Synapse/backend/src/services/resumeTemplates/index.js
- d:/Synapse/backend/src/services/resumeTemplates/researchHigherStudies.js
- d:/Synapse/backend/src/services/resumeTemplates/softwareDeveloper.js
- d:/Synapse/frontend/src/pages/career/ResumeBuilderList.jsx
- d:/Synapse/frontend/src/pages/career/ResumeEditor.jsx
- d:/Synapse/frontend/src/pages/career/ResumeIntelligence.jsx

Internal Implementation & Debugging

When debugging this feature, always check the associated controller and service files first. Ensure MongoDB collections are correctly structured and verify that API requests have the correct JWT tokens. Common issues arise from missing await statements or undefined context variables during AI calls.

Revision Checklist

- ☐ Understand the user journey for Resume Intelligence
- ☐ Review the related models and controllers
- ☐ Identify the specific AI prompts or logic used (if applicable)

Chapter 29: AI Resume Builder

Business Purpose & Workflow

This feature addresses specific student pain points by integrating AI with actionable insights. The workflow typically begins from the user interface, hits the Next.js API routes, passes through middleware, and is processed by the service layer before interacting with MongoDB.

Associated Files

- d:/Synapse/backend/src/controllers/resumeController.js
- d:/Synapse/backend/src/middleware/resumeUpload.js
- d:/Synapse/backend/src/models/Resume.js
- d:/Synapse/backend/src/services/aiContextBuilder.js
- d:/Synapse/backend/src/services/resumeGenerator.js
- d:/Synapse/backend/src/services/resumeParser.js
- d:/Synapse/backend/src/services/resumePdfExporter.js
- d:/Synapse/backend/src/services/resumeRewriteService.js
- d:/Synapse/backend/src/services/resumeSkillGapAnalyzer.js
- d:/Synapse/backend/src/services/resumeTemplates/atsClassic.js

- d:/Synapse/backend/src/services/resumeTemplates/index.js
- d:/Synapse/backend/src/services/resumeTemplates/researchHigherStudies.js
- d:/Synapse/backend/src/services/resumeTemplates/softwareDeveloper.js
- d:/Synapse/frontend/src/pages/career/ResumeBuilderList.jsx
- d:/Synapse/frontend/src/pages/career/ResumeEditor.jsx
- d:/Synapse/frontend/src/pages/career/ResumeIntelligence.jsx

Internal Implementation & Debugging

When debugging this feature, always check the associated controller and service files first. Ensure MongoDB collections are correctly structured and verify that API requests have the correct JWT tokens. Common issues arise from missing await statements or undefined context variables during AI calls.

Revision Checklist

- ☐ Understand the user journey for AI Resume Builder
- ☐ Review the related models and controllers
- ☐ Identify the specific AI prompts or logic used (if applicable)

Conclusion

This manual serves as your foundational knowledge base for defending the project in viva sessions and placement interviews. Remember to connect every technical decision back to the business value it provides to the student.